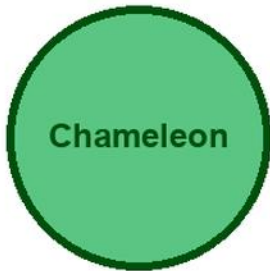


KITABU QUEST

Form I

BIOLOGY



Chameleon

Cover scene

students observing living things safely with a chameleon mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

I

Title Page

KITABU QUEST Form I Biology

Tanzania TIE-BASIC learner book

Mascot: chameleon

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Tanzania learners.

How To Use This Book

This book helps Form I learners study Biology one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use labelled diagrams, safe observations, evidence, and explanations about living things.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Biology As A Science And Laboratory Safety
2. Cells Structure And Microscopes
3. Classification Of Living Things
4. Nutrition In Plants And Animals
5. Respiration And Gas Exchange Basics
6. Movement Support And Coordination
7. Health Hygiene And Disease Prevention
8. Form I Biology Investigation

As you finish each topic, return to the success criteria and correct one answer before moving on.

Biology As A Science And Laboratory Safety: Lesson Opener

Learning focus: Biology As A Science And Laboratory Safety

Country link: a safe Tanzania observation, model, data table, or investigation for Biology As A Science And Laboratory Safety.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain biology as a science and laboratory safety using evidence

Inquiry questions:

- How can biology as a science and laboratory safety help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Biology As A Science And Laboratory Safety: Learn

Biology As A Science And Laboratory Safety begins with safe behaviour and accurate measurement.

Core idea:

- Physics uses measurements such as length, mass, time, temperature, and volume.
- A measurement is useful only when the number and unit are both written.
- Safety rules protect learners, equipment, and results.
- Read scales at eye level and record values immediately.

Local link: in Tanzania, careful measurement helps in health, building, farming, weather records, transport, and school laboratory work.

Key words:

- Biology As A Science And Laboratory Safety
- Biology
- Science
- Laboratory
- Safety
- concept
- observation
- evidence

Biology As A Science And Laboratory Safety: Worked Example

Investigation model for Biology As A Science And Laboratory Safety: Measure the length of a desk safely.

1. Choose a metre rule or tape measure.
2. Place zero at one end of the desk.
3. Read the scale at eye level.
4. Record the value with the unit, for example 120 cm.
5. Repeat once and compare readings.

Conclusion: a useful measurement includes a number, a unit, and a careful method.

Biology As A Science And Laboratory Safety: Vocabulary And Study Skill

Use these words and study moves before you practise Biology As A Science And Laboratory Safety. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Biology As A Science And Laboratory Safety
- Biology
- Science
- Laboratory
- Safety
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Biology As A Science And Laboratory Safety without a clear example, method, or evidence.

Biology As A Science And Laboratory Safety: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Biology As A Science And Laboratory Safety.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Biology As A Science And Laboratory Safety: Practice And Correction

Practice for Biology As A Science And Laboratory Safety:

Main outcome: investigate and explain biology as a science and laboratory safety using evidence.

1. Name three laboratory safety rules.
2. Measure one classroom object and record the value with a unit.
3. Explain one error that can make a measurement unreliable.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Biology As A Science And Laboratory Safety: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Biology As A Science And Laboratory Safety.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Biology As A Science And Laboratory Safety: Outcome Check

Outcome check for Biology As A Science And Laboratory Safety: Investigate and explain biology as a science and laboratory safety using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Cells Structure And Microscopes: Lesson Opener

Learning focus: Cells Structure And Microscopes

Country link: a safe Tanzania observation, model, data table, or investigation for Cells Structure And Microscopes.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain cells structure and microscopes using evidence

Inquiry questions:

- How can cells structure and microscopes help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Cells Structure And Microscopes: Learn

Cells Structure And Microscopes teaches useful computer work through exact steps, not guessing.

Core idea:

- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:

- Cells Structure And Microscopes
- Cells
- Structure
- Microscopes
- concept
- observation
- evidence
- safety

Cells Structure And Microscopes: Worked Example

Worked example for Cells Structure And Microscopes: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Cells Structure And Microscopes: Vocabulary And Study Skill

Use these words and study moves before you practise Cells Structure And Microscopes. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Cells Structure And Microscopes
- Cells
- Structure
- Microscopes
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Cells Structure And Microscopes without a clear example, method, or evidence.

Cells Structure And Microscopes: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Cells Structure And Microscopes.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Cells Structure And Microscopes: Practice And Correction

Practice for Cells Structure And Microscopes:

Main outcome: investigate and explain cells structure and microscopes using evidence.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Cells Structure And Microscopes: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Cells Structure And Microscopes.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Cells Structure And Microscopes: Outcome Check

Outcome check for Cells Structure And Microscopes: Investigate and explain cells structure and microscopes using evidence.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Classification Of Living Things: Lesson Opener

Learning focus: Classification Of Living Things

Country link: a safe Tanzania observation, model, data table, or investigation for Classification Of Living Things.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain classification of living things using evidence

Inquiry questions:

- How can classification of living things help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Classification Of Living Things: Learn

Classification Of Living Things is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Classification Of Living Things
- Classification
- Living
- Things
- concept

Classification Of Living Things: Worked Example

Investigation model for Classification Of Living Things: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Classification Of Living Things: Vocabulary And Study Skill

Use these words and study moves before you practise Classification Of Living Things. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Classification Of Living Things
- Classification
- Living
- Things
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Classification Of Living Things without a clear example, method, or evidence.

Classification Of Living Things: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Classification Of Living Things.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Classification Of Living Things: Practice And Correction

Practice for Classification Of Living Things:

Main outcome: investigate and explain classification of living things using evidence.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Classification Of Living Things: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Classification Of Living Things.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Classification Of Living Things: Outcome Check

Outcome check for Classification Of Living Things: Investigate and explain classification of living things using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Nutrition In Plants And Animals: Lesson Opener

Learning focus: Nutrition In Plants And Animals

Country link: a safe Tanzania observation, model, data table, or investigation for Nutrition In Plants And Animals.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain nutrition in plants and animals using evidence

Inquiry questions:

- How can nutrition in plants and animals help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Nutrition In Plants And Animals: Learn

Nutrition In Plants And Animals is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Nutrition In Plants And Animals
- Nutrition
- Plants
- Animals
- concept
- observation
- evidence
- safety

Nutrition In Plants And Animals: Worked Example

Investigation model for Nutrition In Plants And Animals: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Nutrition In Plants And Animals: Vocabulary And Study Skill

Use these words and study moves before you practise Nutrition In Plants And Animals. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Nutrition In Plants And Animals
- Nutrition
- Plants
- Animals
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Nutrition In Plants And Animals without a clear example, method, or evidence.

Nutrition In Plants And Animals: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Nutrition In Plants And Animals.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Nutrition In Plants And Animals: Practice And Correction

Practice for Nutrition In Plants And Animals:

Main outcome: investigate and explain nutrition in plants and animals using evidence.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Nutrition In Plants And Animals: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Nutrition In Plants And Animals.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Nutrition In Plants And Animals: Outcome Check

Outcome check for Nutrition In Plants And Animals: Investigate and explain nutrition in plants and animals using evidence.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Respiration And Gas Exchange Basics: Lesson Opener

Learning focus: Respiration And Gas Exchange Basics

Country link: a safe Tanzania observation, model, data table, or investigation for Respiration And Gas Exchange Basics.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain respiration and gas exchange basics using evidence

Inquiry questions:

- How can respiration and gas exchange basics help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Respiration And Gas Exchange Basics: Learn

Respiration And Gas Exchange Basics explains particles in solids, liquids, and gases.

Core idea:

- Matter is made of tiny particles.
- In solids, particles are close and vibrate in fixed positions.
- In liquids, particles are close but can slide past each other.
- In gases, particles are far apart and move freely.
- Heating usually makes particles move faster.

Key words:

- Respiration And Gas Exchange Basics
- Respiration
- Exchange
- Basics
- concept
- observation
- evidence
- safety

Respiration And Gas Exchange Basics: Worked Example

Model for Respiration And Gas Exchange Basics: Compare particles in solids, liquids, and gases.

1. Draw solid particles close together in a fixed pattern.
2. Draw liquid particles close but uneven, able to slide.
3. Draw gas particles far apart.
4. Add arrows to show particle movement.

Conclusion: particle arrangement explains shape, flow, and expansion.

Respiration And Gas Exchange Basics: Vocabulary And Study Skill

Use these words and study moves before you practise Respiration And Gas Exchange Basics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Respiration And Gas Exchange Basics
- Respiration
- Exchange
- Basics
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Respiration And Gas Exchange Basics without a clear example, method, or evidence.

Respiration And Gas Exchange Basics: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Respiration And Gas Exchange Basics.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Respiration And Gas Exchange Basics: Practice And Correction

Practice for Respiration And Gas Exchange Basics:

Main outcome: investigate and explain respiration and gas exchange basics using evidence.

1. Draw particles in a solid, liquid, and gas.
2. Explain why gases fill containers.
3. State what heating does to particle movement.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Respiration And Gas Exchange Basics: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Respiration And Gas Exchange Basics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Respiration And Gas Exchange Basics: Outcome Check

Outcome check for Respiration And Gas Exchange Basics: Investigate and explain respiration and gas exchange basics using evidence.

1. Compare particles in a solid and a gas.
2. Draw a simple particle diagram.
3. Explain what heating changes.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Movement Support And Coordination: Lesson Opener

Learning focus: Movement Support And Coordination

Country link: a safe Tanzania observation, model, data table, or investigation for Movement Support And Coordination.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain movement support and coordination using evidence

Inquiry questions:

- How can movement support and coordination help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Movement Support And Coordination: Learn

Movement Support And Coordination explains how muscles help the body move and stay active.

Core idea:

- Muscles pull on bones to create movement at joints.
- Muscles usually work in pairs: one contracts while the other relaxes.
- Exercise, rest, clean food, and safe posture help muscles remain healthy.
- Sudden heavy work, poor warm-up, and unsafe movement can injure muscles.
- A labelled body diagram helps learners connect muscle action with daily movement.

Key words:

- Movement Support And Coordination
- Movement
- Support
- Coordination
- concept
- observation
- evidence
- safety

Movement Support And Coordination: Worked Example

Model for Movement Support And Coordination: Explain how the arm bends.

1. Hold one arm out and bend it slowly.
2. Feel the upper-arm muscle shorten as the elbow bends.
3. Straighten the arm and notice the movement changes.
4. Draw a simple arm diagram with bone, joint, and muscle labels.
5. Add one safety note about warm-up or avoiding sudden heavy lifting.

Conclusion: muscles pull on bones at joints to make movement.

Movement Support And Coordination: Vocabulary And Study Skill

Use these words and study moves before you practise Movement Support And Coordination. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Movement Support And Coordination
- Movement
- Support
- Coordination
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Movement Support And Coordination without a clear example, method, or evidence.

Movement Support And Coordination: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Movement Support And Coordination.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Movement Support And Coordination: Practice And Correction

Practice for Movement Support And Coordination:

Main outcome: investigate and explain movement support and coordination using evidence.

1. Draw and label a simple joint and muscle action.
2. Explain how a muscle helps movement by pulling.
3. List two habits that protect muscles during work or exercise.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Movement Support And Coordination: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Movement Support And Coordination.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Movement Support And Coordination: Outcome Check

Outcome check for Movement Support And Coordination: Investigate and explain movement support and coordination using evidence.

1. State how muscles create movement.
2. Name one joint or body movement example.
3. Write one habit that keeps muscles healthy.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Health Hygiene And Disease Prevention: Lesson Opener

Learning focus: Health Hygiene And Disease Prevention

Country link: a safe Tanzania observation, model, data table, or investigation for Health Hygiene And Disease Prevention.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain health hygiene and disease prevention using evidence

Inquiry questions:

- How can health hygiene and disease prevention help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Health Hygiene And Disease Prevention: Learn

Health Hygiene And Disease Prevention is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Health Hygiene And Disease Prevention
- Health
- Hygiene
- Disease
- Prevention
- concept
- observation
- evidence

Health Hygiene And Disease Prevention: Worked Example

Investigation model for Health Hygiene And Disease Prevention: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Health Hygiene And Disease Prevention: Vocabulary And Study Skill

Use these words and study moves before you practise Health Hygiene And Disease Prevention. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Health Hygiene And Disease Prevention
- Health
- Hygiene
- Disease
- Prevention
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Health Hygiene And Disease Prevention without a clear example, method, or evidence.

Health Hygiene And Disease Prevention: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Health Hygiene And Disease Prevention.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Health Hygiene And Disease Prevention: Practice And Correction

Practice for Health Hygiene And Disease Prevention:

Main outcome: investigate and explain health hygiene and disease prevention using evidence.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Health Hygiene And Disease Prevention: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Health Hygiene And Disease Prevention.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Health Hygiene And Disease Prevention: Outcome Check

Outcome check for Health Hygiene And Disease Prevention: Investigate and explain health hygiene and disease prevention using evidence.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Form I Biology Investigation: Lesson Opener

Learning focus: Form I Biology Investigation

Country link: a safe Tanzania observation, model, data table, or investigation for Form I Biology Investigation.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain form i biology investigation using evidence

Inquiry questions:

- How can form i biology investigation help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Form I Biology Investigation: Learn

Form I Biology Investigation is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Form I Biology Investigation
- Form
- Biology
- Investigation
- concept
- observation
- evidence
- safety

Form I Biology Investigation: Worked Example

Investigation model for Form I Biology Investigation: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Form I Biology Investigation: Vocabulary And Study Skill

Use these words and study moves before you practise Form I Biology Investigation. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Form I Biology Investigation
- Form
- Biology
- Investigation
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Form I Biology Investigation without a clear example, method, or evidence.

Form I Biology Investigation: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Form I Biology Investigation.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Form I Biology Investigation: Practice And Correction

Practice for Form I Biology Investigation:

Main outcome: investigate and explain form i biology investigation using evidence.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Form I Biology Investigation: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Form I Biology Investigation.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Form I Biology Investigation: Outcome Check

Outcome check for Form I Biology Investigation: Investigate and explain form i biology investigation using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Biology As A Science And Laboratory Safety: Review Clinic 1

Use this review clinic to strengthen Biology without copying earlier answers.

1. Choose one idea from Biology As A Science And Laboratory Safety that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Cells Structure And Microscopes: Review Clinic 2

Use this review clinic to strengthen Biology without copying earlier answers.

1. Choose one idea from Cells Structure And Microscopes that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Classification Of Living Things: Review Clinic 3

Use this review clinic to strengthen Biology without copying earlier answers.

1. Choose one idea from Classification Of Living Things that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Nutrition In Plants And Animals: Review Clinic 4

Use this review clinic to strengthen Biology without copying earlier answers.

1. Choose one idea from Nutrition In Plants And Animals that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Biology As A Science And Laboratory Safety: Application Workshop 1

Application workshop 1 for Biology As A Science And Laboratory Safety.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Cells Structure And Microscopes: Application Workshop 2

Application workshop 2 for Cells Structure And Microscopes.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Classification Of Living Things: Application Workshop 3

Application workshop 3 for Classification Of Living Things.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Nutrition In Plants And Animals: Application Workshop 4

Application workshop 4 for Nutrition In Plants And Animals.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Respiration And Gas Exchange Basics: Application Workshop 5

Application workshop 5 for Respiration And Gas Exchange Basics.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Movement Support And Coordination: Application Workshop 6

Application workshop 6 for Movement Support And Coordination.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Health Hygiene And Disease Prevention: Application Workshop 7

Application workshop 7 for Health Hygiene And Disease Prevention.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Form I Biology Investigation: Application Workshop 8

Application workshop 8 for Form I Biology Investigation.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Biology As A Science And Laboratory Safety: Application Workshop 9

Application workshop 9 for Biology As A Science And Laboratory Safety.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Cells Structure And Microscopes: Application Workshop 10

Application workshop 10 for Cells Structure And Microscopes.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Glossary

Use this glossary to revise important words in Biology.

- Biology As A Science And Laboratory Safety: Explain this term using an example from the book, your school, or your community.
- Cells Structure And Microscopes: Explain this term using an example from the book, your school, or your community.
- Classification Of Living Things: Explain this term using an example from the book, your school, or your community.
- Nutrition In Plants And Animals: Explain this term using an example from the book, your school, or your community.
- Respiration And Gas Exchange Basics: Explain this term using an example from the book, your school, or your community.
- Movement Support And Coordination: Explain this term using an example from the book, your school, or your community.
- Health Hygiene And Disease Prevention: Explain this term using an example from the book, your school, or your community.
- Form I Biology Investigation: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Biology answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Biology that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.